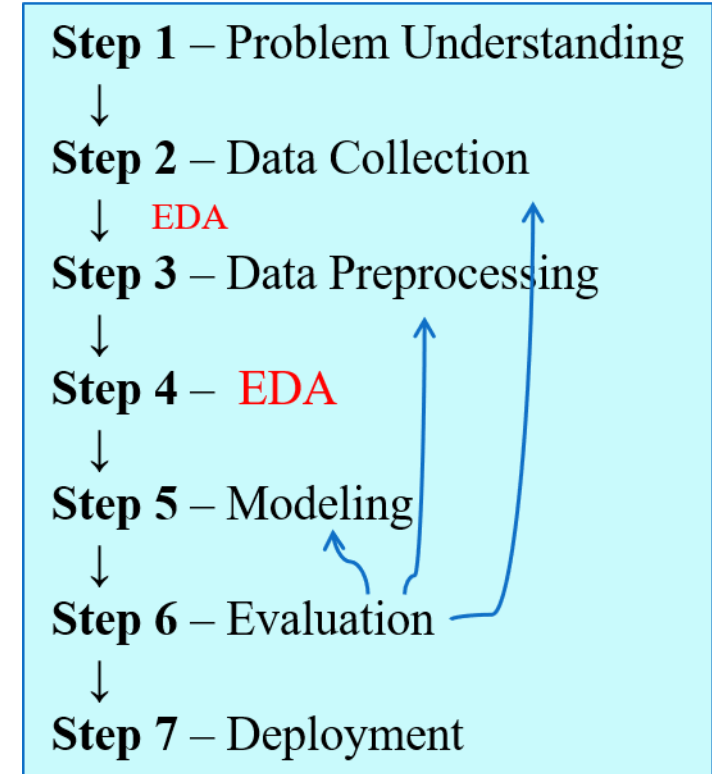


Data Preprocessing

- Raw data is often noisy, inconsistent, and incomplete.
- Preprocessing improves data quality and model performance.
- Essential step before data analysis or machine learning.

Data preprocessing **transforms raw data into a clean and usable format** for analysis and machine learning models.

It's a crucial first step that involves cleaning, transforming, integrating, and reducing data to improve **quality and efficiency**.



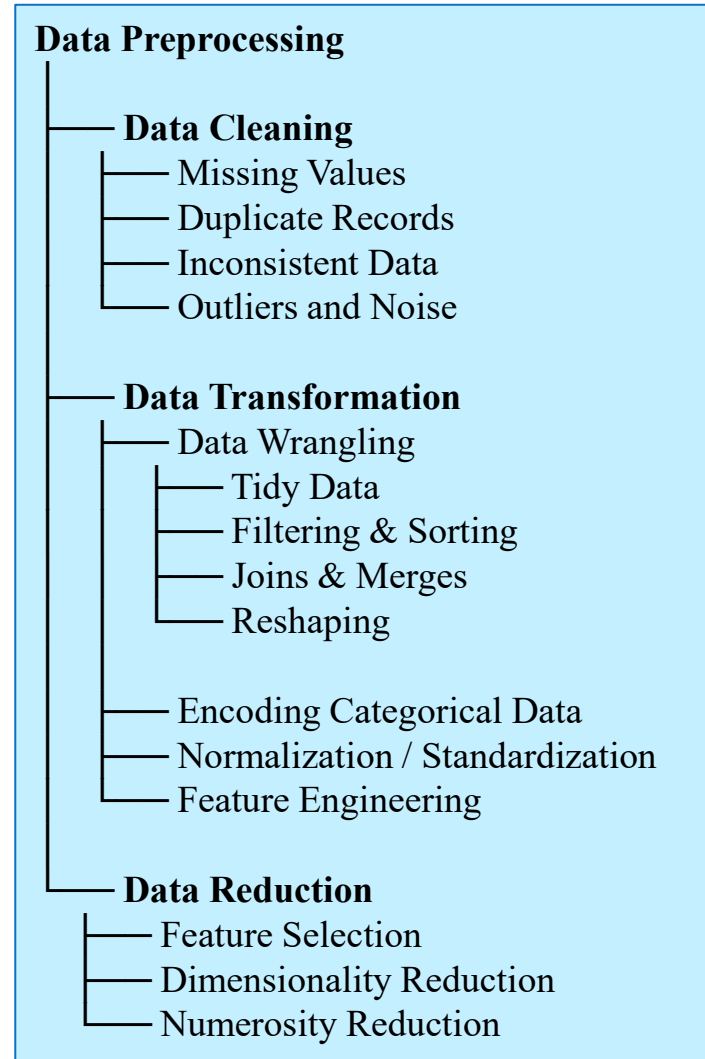
Data Preprocessing

The main steps are

Data Cleaning: This step focuses on **fixing issues within the data** to improve its quality. Think of it as tidying up your dataset.

Data Transformation: This step involves **modifying the data's structure or format** to make it more suitable for machine learning algorithms.

Data Reduction: This step aims to **reduce the size of the dataset** without sacrificing important information, making the analysis faster and more efficient.



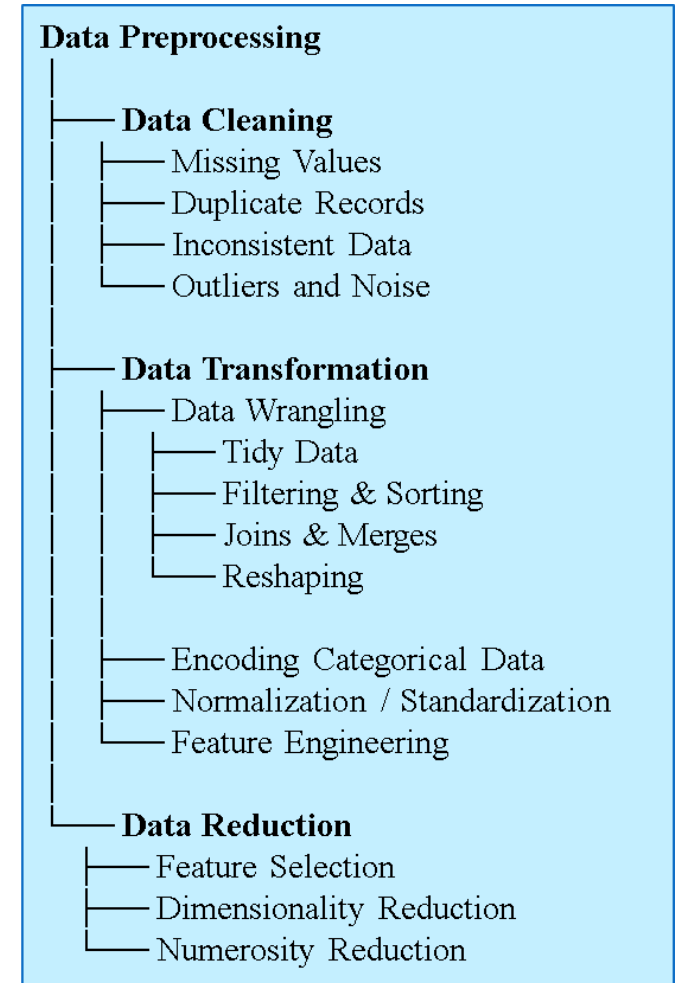
Data Preprocessing : Data Cleaning

Handling Missing Values

Handling missing values is a crucial data cleaning step because most machine learning algorithms **cannot work with empty or null data points**.

Choosing the right method depends on the **nature of the data and the amount of missing information**.

Let's use a simple example dataset to discuss the **common techniques**.



Data Preprocessing : Data Cleaning

Handling Missing Values

Example Dataset: Employee Info

Imagine we have the following data about employees. The **NaN** values represent missing data.

Employee ID	Age	Experience (Years)	Salary
101	25	2	50,000
102	30	NaN	65,000
103	45	20	120,000
104	28	5	NaN
105	50	25	140,000

Data Preprocessing : Data Cleaning

Handling Missing Values

1. Deletion Methods

The simplest approach is to just remove the missing data.

- **Listwise Deletion:** This involves deleting the entire row that contains a missing value. In our example, we would remove the rows for Employee ID 102 and 104.

Employee ID	Age	Experience (Years)	Salary
101	25	2	50,000
102	30	NaN	65,000
103	45	20	120,000
104	28	5	NaN
105	50	25	140,000



Employee ID	Age	Experience (Years)	Salary
101	25	2	50,000
103	45	20	120,000
105	50	25	140,000

Data Preprocessing : Data Cleaning

Handling Missing Values

1. Deletion Methods

Pros: Easy and fast.

Cons:

- You lose a lot of data, which can lead to a biased model if the missing values aren't random.
- This is only recommended if you have a very large dataset and only a few rows with missing values.

Employee ID	Age	Experience (Years)	Salary
101	25	2	50,000
102	30	NaN	65,000
103	45	20	120,000
104	28	5	NaN
105	50	25	140,000



Employee ID	Age	Experience (Years)	Salary
101	25	2	50,000
103	45	20	120,000
105	50	25	140,000

Data Preprocessing : Data Cleaning

Handling Missing Values

2. Imputation with a Single Value

Imputation means filling in the missing values. The most common way is to use a statistical measure from the column.

- **Mean/Median Imputation:** You can replace missing numerical data with the mean (average) or median (middle value) of the column. **The median is often preferred because it's less sensitive to outliers.**
- **Mode Imputation:** For categorical data (e.g., a 'Department' column), you'd use the mode (the most frequent value).

Let's apply this to our original dataset.

Data Preprocessing : Data Cleaning

Handling Missing Values

2. Imputation with a Single Value

Step 1: Calculate the mean for 'Experience' and 'Salary'.

Employee ID	Age	Experience (Years)	Salary
101	25	2	50,000
102	30	NaN	65,000
103	45	20	120,000
104	28	5	NaN
105	50	25	140,000

Known 'Experience' values: 2, 20, 5, 25.

Mean Experience = $(2+20+5+25)/4=13$ years.

Known 'Salary' values: 50000, 65000, 120000, 140000.

Mean Salary = $(50000+65000+120000+140000)/4=93,750$.

Data Preprocessing : Data Cleaning

Handling Missing Values

2. Imputation with a Single Value

Step 2: Fill in the NaN values.

- The missing 'Experience' for employee 102 becomes 13.
- The missing 'Salary' for employee 104 becomes 93,750.

Pros:

Simple to implement and retains all data rows.

Cons:

Can reduce the natural variance in the data and weaken the relationships between variables, as it doesn't account for correlations.

For instance, a 28-year-old is now assigned the average salary, which might not be realistic.

Employee ID	Age	Experience (Years)	Salary
101	25	2	50,000
102	30	NaN	65,000
103	45	20	120,000
104	28	5	NaN
105	50	25	140,000



Employee ID	Age	Experience (Years)	Salary
101	25	2	50,000
102	30	13	65,000
103	45	20	120,000
104	28	5	93,750
105	50	25	140,000

Data Preprocessing : Data Cleaning

Handling Missing Values

3. Advanced Imputation Method

More sophisticated techniques use other features to predict the missing values.

- **Regression Imputation:** This method builds a regression model to predict the missing value. For example, we could create a model to predict 'Salary' based on 'Age' and 'Experience'. We would train this model on the complete rows (101, 103, 105) and then use it to predict the salary for employee 104.
- **K-Nearest Neighbors (KNN) Imputation:** This method finds the 'k' most similar data points (neighbors) to the one with the missing value and then imputes the missing value based on the average of its neighbors.

Pros: Usually more accurate than simple imputation and better at preserving relationships between variables.

Cons: Much more complex to implement and computationally expensive.

Data Preprocessing : Data Cleaning

Handling Duplicate Records

Duplicate records occur when the same observation appears more than once in a dataset.

Why Are Duplicate Records a Problem?

- Duplicate records can:
- Inflate dataset size
- Produce incorrect statistics
- Bias machine learning models
- Lead to incorrect business decisions

StudentID	Name	CGPA
101	Rahim	3.75
102	Karim	3.40
102	Karim	3.40
103	Nabila	3.90



StudentID	Name	CGPA
101	Rahim	3.75
102	Karim	3.40
103	Nabila	3.90

Data Preprocessing

Data Cleaning

- Missing Values
- Duplicate Records
- Inconsistent Data
- Outliers and Noise

Data Transformation

- Data Wrangling
 - Tidy Data
 - Filtering & Sorting
 - Joins & Merges
 - Reshaping

- Encoding Categorical Data
- Normalization / Standardization
- Feature Engineering

Data Reduction

- Feature Selection
- Dimensionality Reduction
- Numerosity Reduction

Data Preprocessing : Data Cleaning

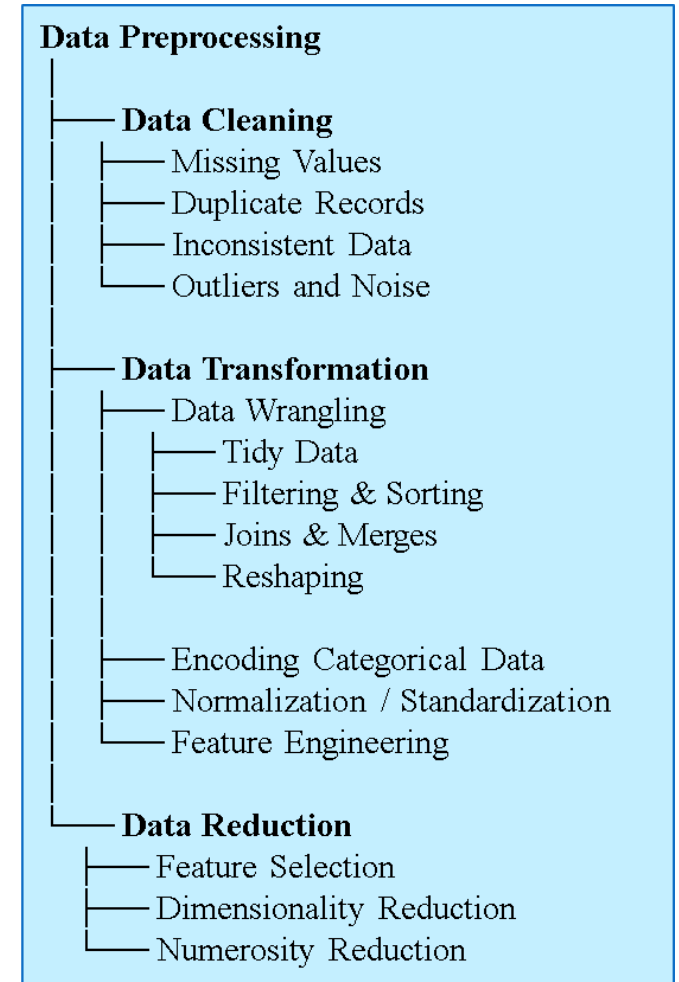
Correcting Inaccurate or Inconsistent Data

Correcting inaccurate or inconsistent data is the process of identifying and **fixing errors, contradictions, and structural issues** to ensure your dataset is reliable and uniform.

Think of it as proofreading your data to make sure every entry is accurate and follows the same set of rules.

This process is vital because inconsistencies can lead to incorrect analysis, biased models, and flawed conclusions.

Next is how to do it, broken down by common types of errors.



Data Preprocessing : Data Cleaning

Correcting Inaccurate or Inconsistent Data

1. Correcting Inconsistent Categorical Data

This happens when the same category is represented in multiple ways.

The Problem: A 'Country' column might contain "USA," "U.S.A.," "United States," and "America." A data analysis tool would treat these as four different countries.

How to Fix It (Standardization):

- *Explore Unique Values:* First, identify all the unique entries in the column
- *Define a Standard:* Decide on a single, standard representation.
- *Create a Mapping:* Create a dictionary to map the variations to your chosen standard. Use the map to replace the inconsistent values.

Data Preprocessing : Data Cleaning

Correcting Inaccurate or Inconsistent Data

2. Fixing Structural Errors

These are typos, unwanted characters, or inconsistent capitalization that affect data quality.

The Problem: A 'Name' column might have entries like " John Smith ", "john smith", or "John_Smith".

How to Fix It (Cleaning and Formatting):

- *Trim Whitespace:* Remove any leading or trailing spaces. Most programming languages have a trim() or strip() function.
- *Standardize Case:* Convert all text to a single case (usually lowercase) to ensure consistency. Use a lower() function.
- *Remove Unwanted Characters:* Use a replace() function to remove special characters like underscores or dashes if they don't belong.

Data Preprocessing : Data Cleaning

Correcting Inaccurate or Inconsistent Data

3. Resolving Logical Errors and Contradictions

This involves finding data points that are logically impossible or contradict other information in the dataset.

The Problem:

- A person's Date of Death is earlier than their Date of Birth.
- An 'Age' column shows 15, but the 'Employment Status' column says Full-Time Employee.
- An Order_Date is later than the Shipping_Date.

Data Preprocessing : Data Cleaning

Correcting Inaccurate or Inconsistent Data

3. Resolving Logical Errors and Contradictions

How to Fix It (Validation Rules):

Define Rules: Establish logical rules that your data must follow (e.g., Age must be greater than 18 for Employment Status == 'Full-Time Employee').

Flag Violations: Write a script to check each row against your rules and flag the ones that fail.

Correct Manually or Programmatically: For some errors, you might need to **investigate the original source** to find the correct data. For others, you might set the problematic entry to **NaN** (missing) and handle it later. If a Date of Birth is "1992" and Age is "20", you could write a rule to **recalculate** the age based on the current date.

Data Preprocessing : Data Cleaning

Correcting Inaccurate or Inconsistent Data

4. Standardizing Units and Formats

This ensures all numerical data and dates are comparable.

The Problem:

- A 'Weight' column contains values in both pounds (lbs) and kilograms (kg).
- A 'Date' column has multiple formats like "07-25-2025," "25/Jul/2025."

How to Fix It (Conversion and Parsing):

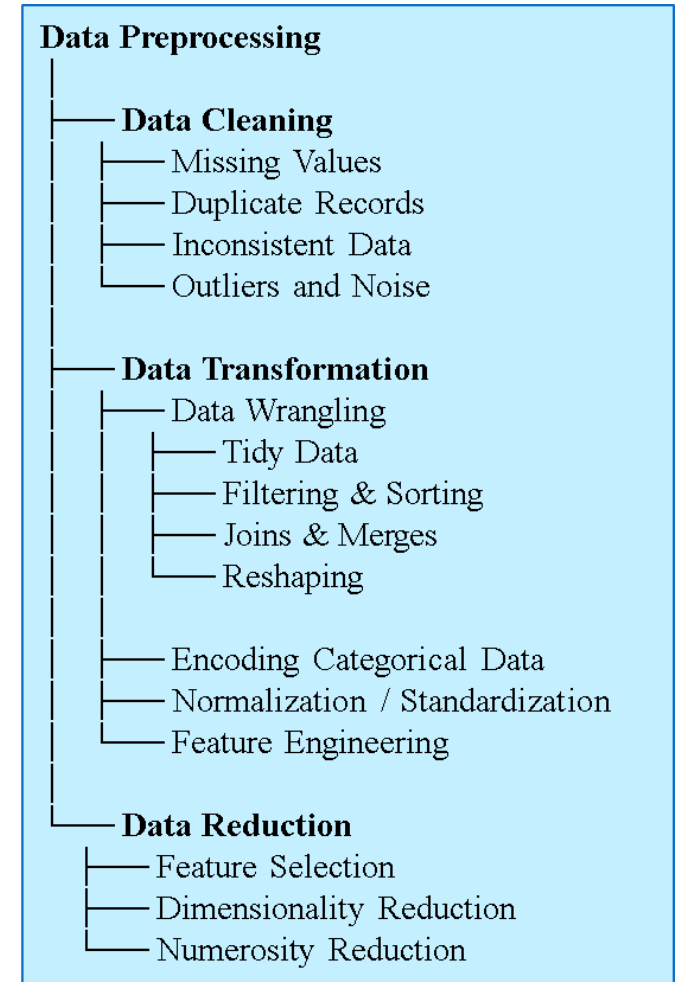
- *For Numerical Units:* Identify the different units present. Choose a standard unit (e.g., kg) and apply a conversion formula to all non-standard entries (e.g., $\text{weight_in_kg} = \text{weight_in_lbs} * 0.453592$).
- *For Dates:* Use a robust date-parsing function to automatically recognize and convert all date formats into a single, standard format, like YYYY-MM-DD.

Data Preprocessing : Data Cleaning

Handling Outliers and Noise

Handling outliers and noise is a critical data cleaning step to prevent skewed results and improve model accuracy. While related, they are different:

- Outliers are legitimate data points that are unusually far from the rest of the data. Think of the net worth of a billionaire in a sample of regular citizens.
- Noise is random error or variance in the data, like a slight fluctuation in a sensor reading.



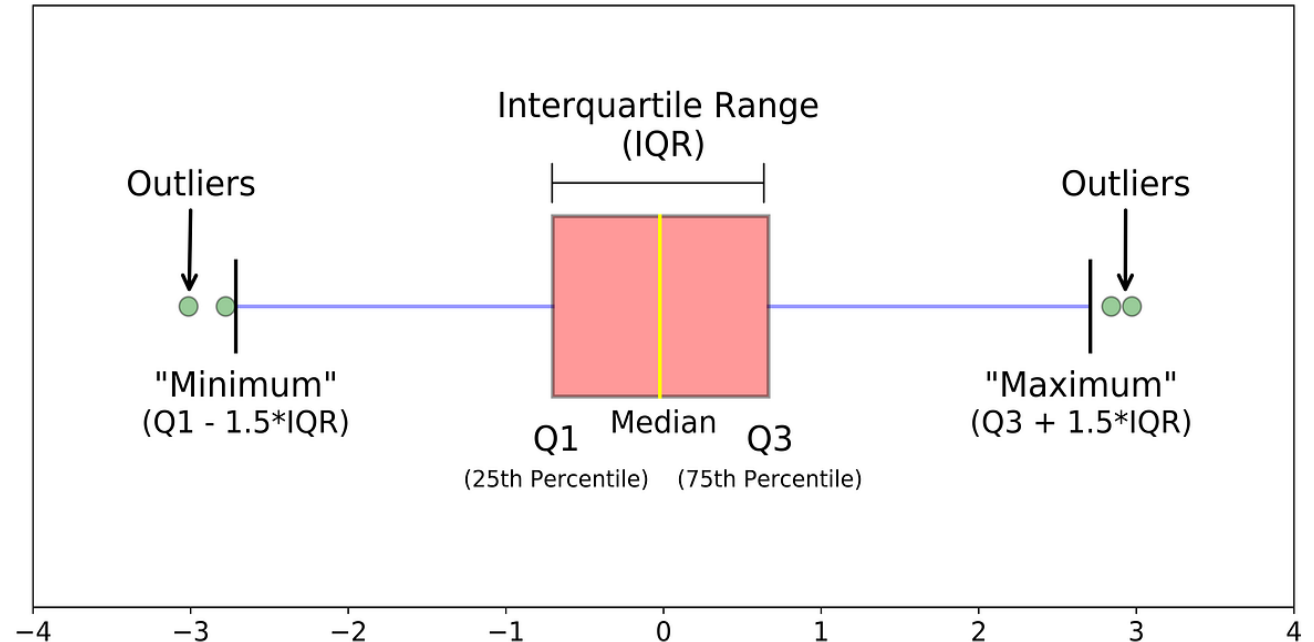
Data Preprocessing : Data Cleaning

Outliers Detection

The first step is to find the outliers, and then you can decide what to do with them.

Visual Method (Box Plots):

This is the easiest way to "see" outliers. A box plot visualizes the distribution of data. Any data point that falls outside the "whiskers" of the plot is typically considered an outlier.



Data Preprocessing : Data Cleaning

Outliers Detection

Statistical Method (Interquartile Range - IQR):

The IQR method provides a numerical way to identify outliers.

- Calculate the first quartile (Q1, the 25th percentile) and the third quartile (Q3, the 75th percentile).
- Calculate the IQR: $IQR = Q3 - Q1$.
- Define the outlier boundaries. A data point is considered an outlier if it falls below $Q1 - 1.5 \text{ times IQR}$ or above $Q3 + 1.5 \text{ times IQR}$.

Z-Score Method:

The Z-score tells you how many standard deviations a data point is from the mean. A common rule of thumb is that any point with a Z-score greater than 3 or less than -3 is an outlier.

Data Preprocessing : Data Cleaning

Outliers Handling

Method	Description	When to use
Remove	Delete outlier rows	Small number; known data entry errors
Cap / Winsorize	Replace outliers with nearest boundary value	Preserve sample size; mild outliers
Transform	Log, Box-Cox, etc.	Reduce skewness; keep all data
Treat separately	Keep as a separate category	Meaningful outliers (e.g., fraud, extreme events)

Data Preprocessing : Data Cleaning

Handling Noise

Handling noise involves using "smoothing" techniques to remove random fluctuations and reveal the underlying trend.

1. Binning:

This method groups a range of continuous values into "bins" or intervals. After creating the bins, you can smooth the data by replacing every value in a bin with the bin's mean, median, or boundary value.

- Example: Imagine the sorted price data: [3, 5, 8, 15, 16, 17, 25, 26, 30].
- You could create 3 bins: [3, 5, 8], [15, 16, 17], and [25, 26, 30].
- If you smooth by the mean, the data becomes [5.3, 5.3, 5.3], [16, 16, 16], and [27, 27, 27].

Data Preprocessing : Data Cleaning

Handling Noise

2. Regression:

You can fit a regression line (or curve) to the data. The data points on this line are considered the "smoothed," noise-free values. The original data points are then replaced with their corresponding values on the regression line.

3. Moving Averages:

This is very common for time-series data (like stock prices or temperature readings). It smooths the data by calculating the average of a "sliding window" of a fixed size. For example, a 7-day moving average would replace each day's value with the average of that day, the 3 previous days, and the 3 following days. This smooths out daily fluctuations and highlights the longer-term trend.

Data Preprocessing : Data Transformation

Data Wrangling

Data Wrangling (also called Data Munging) is the process of transforming raw data into a clean, structured, and analysis-ready format.

Goal

- Organize data
- Restructure data
- Combine data from multiple sources
- Prepare data for analysis and machine learning

Data Preprocessing

Data Cleaning

- Missing Values
- Duplicate Records
- Inconsistent Data
- Outliers and Noise

Data Transformation

- Data Wrangling
 - Tidy Data
 - Filtering & Sorting
 - Joins & Merges
 - Reshaping
- Encoding Categorical Data
- Normalization / Standardization
- Feature Engineering

Data Reduction

- Feature Selection
- Dimensionality Reduction
- Numerosity Reduction

Data Preprocessing : Data Transformation

Tidy Data

Tidy Data is a standardized way of organizing data where:

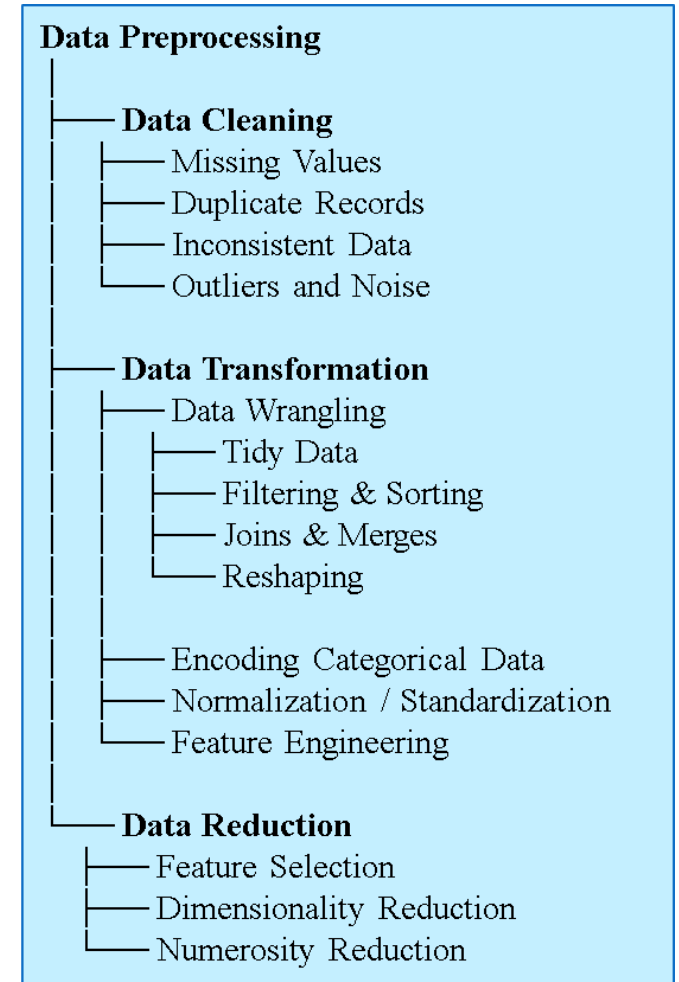
- Each variable forms a column
- Each observation forms a row
- Each value occupies a single cell

These principles were popularized by Hadley Wickham.

Most data analysis and visualization tools work best when data follows a consistent structure.

Benefits:

- Easier analysis
- Easier visualization
- Easier data manipulation



Data Preprocessing : Data Transformation

Common Data Wrangling Tasks:

Filtering: Select rows that satisfy specific conditions.

Student	CGPA
Rahim	3.8
Karim	3.2
Nabila	3.9

Filter: $\text{CGPA} > 3.5$

Student	CGPA
Rahim	3.8
Nabila	3.9

Data Preprocessing

Data Cleaning

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- Feature Selection
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- Numerosity Reduction

Data Preprocessing : Data Transformation

Common Data Wrangling Tasks

Sorting: Arrange data in ascending or descending order.

Student	CGPA
Rahim	3.8
Karim	3.2
Nabila	3.9

After sorting (descending):

Student	CGPA
Nabila	3.9
Rahim	3.8
Karim	3.2

Data Preprocessing

Data Cleaning

- Missing Values
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- Numerosity Reduction

Data Preprocessing : Data Transformation

Common Data Wrangling Tasks:

Joins and Merges: Combine information from multiple datasets.

ID	Name
1	Rahim
2	Karim

ID	CGPA
1	3.8
2	3.5

After Merge

ID	Name	CGPA
1	Rahim	3.8
2	Karim	3.5

Data Preprocessing

Data Cleaning

- Missing Values
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- Numerosity Reduction

Data Preprocessing : Data Transformation

Common Data Wrangling Tasks:

Reshaping Data: Change the structure of a dataset.

Student	Quiz1	Quiz2
Rahim	80	85
Karim	75	88

Student	Quiz	Marks
Rahim	Quiz1	80
Rahim	Quiz2	85
Karim	Quiz1	75
Karim	Quiz2	88

Data Preprocessing

Data Cleaning

- Missing Values
- Duplicate Records
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Data Transformation

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Data Preprocessing : Data Transformation

Encoding Categorical Data

Encoding categorical data is the process of converting text-based categories into numbers so that machine learning algorithms can process them.

Let's use a simple example. Imagine you have survey data with a "Favorite Fruit" column.

A machine learning model can't understand "Apple" or "Orange." We need to encode these labels into numbers. Here are the two most common ways to do that.

Person ID	Favorite Fruit
1	Apple
2	Orange
3	Banana
4	Apple
5	Banana

Data Preprocessing

Data Cleaning

- Missing Values
- Duplicate Records
- Inconsistent Data
- Outliers and Noise

Data Transformation

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Data Preprocessing : Data Transformation

1. Label Encoding

Important Note:

- Be careful using this for non-ordered (nominal) data.
- Label Encoding creates an artificial order (Banana (2) > Orange (1)), which can mislead some models into thinking there's a ranking when there isn't one.

Person ID	Favorite Fruit
1	Apple
2	Orange
3	Banana
4	Apple
5	Banana



Person ID	Favorite Fruit	Favorite Fruit (Encoded)
1	Apple	0
2	Orange	1
3	Banana	2
4	Apple	0
5	Banana	2

Data Preprocessing : Data Transformation

2. One-Hot Encoding

This is the preferred method for nominal data (categories without a natural order). It creates a new binary (0 or 1) column for each unique category.

How it works:

- Identify all unique categories: "Apple," "Orange," "Banana."
- Create a new column for each category.
- For each row, place a 1 in the column corresponding to its category and 0s in all other new columns.

Person ID	Favorite Fruit
1	Apple
2	Orange
3	Banana
4	Apple
5	Banana



Person ID	Favorite Fruit	Apple	Orange	Banana
1	Apple	1	0	0
2	Orange	0	1	0
3	Banana	0	0	1
4	Apple	1	0	0
5	Banana	0	0	1

Data Preprocessing : Data Transformation

2. One-Hot Encoding

This method effectively represents the categories without creating any false relationships between them, making it a safer and often more effective choice for most machine learning tasks.

Person ID	Favorite Fruit
1	Apple
2	Orange
3	Banana
4	Apple
5	Banana



Person ID	Favorite Fruit	Apple	Orange	Banana
1	Apple	1	0	0
2	Orange	0	1	0
3	Banana	0	0	1
4	Apple	1	0	0
5	Banana	0	0	1

Data Preprocessing : Data Transformation

Normalization and Standardization

Normalization and Standardization are two essential data transformation techniques used to scale numerical features to a common level, but they do so in different ways.

Let's use a simple dataset to illustrate both. Imagine we have data for a small group of people and we want to scale the 'Income' feature.

Person	Age	Income
A	25	50,000
B	30	65,000
C	45	120,000
D	50	140,000
E	35	95,000

Data Preprocessing

Data Cleaning

- Missing Values
- Duplicate Records
- Inconsistent Data
- Outliers and Noise

Data Transformation

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- Numerosity Reduction

Data Preprocessing : Data Transformation

Normalization (Min-Max Scaling)

Normalization and Standardization are two essential data transformation techniques used to scale numerical features to a common level, but they do so in different ways.

Let's use a simple dataset to illustrate both. Imagine we have data for a small group of people and we want to scale the 'Income' feature.

Person	Age	Income
A	25	50,000
B	30	65,000
C	45	120,000
D	50	140,000
E	35	95,000

$$\text{Formula: } X_{norm} = \frac{X - X_{min}}{X_{max} - X_{min}}$$

Example Calculation for 'Income'

Step 1: Find the minimum and maximum values for 'Income'.

$$X_{min} = 50,000$$

$$X_{max} = 140,000$$

Data Preprocessing : Data Transformation

Normalization (Min-Max Scaling)

Step 2: Apply the formula to each 'Income' value.

- Person A (50,000):

$$\frac{50,000 - 50,000}{140,000 - 50,000} = 0$$

- Person B (65,000):

$$\frac{65,000 - 50,000}{140,000 - 50,000} \approx 0.167$$

Similarly,

Person C (120,000): ≈ 0.778

Person D (140,000): $= 1.0$

Person E (95,000): $= 0.5$

Person	Age	Income
A	25	50,000
B	30	65,000
C	45	120,000
D	50	140,000
E	35	95,000



Person	Age	Income
A	25	0
B	30	0.167
C	45	0.778
D	50	1.0
E	35	0.5

Data Preprocessing : Data Transformation

Standardization (Z-Score Scaling)

Goal: To rescale data so it has a mean (μ) of 0 and a standard deviation (σ) of 1. This is useful for algorithms that assume your data follows a **Gaussian (normal) distribution** (e.g., Linear Regression, Logistic Regression).

Person	Age	Income
A	25	50,000
B	30	65,000
C	45	120,000
D	50	140,000
E	35	95,000

Formula: $X_{std} = \frac{X - \mu}{\sigma}$

Example Calculation for 'Income'

'Step 1: Calculate the mean (μ) and standard deviation (σ) for 'Income'.

- Mean (μ):

$$\frac{50000 + 65000 + 120000 + 140000 + 95000}{5} = 94,000$$

Data Preprocessing : Data Transformation

Standardization (Z-Score Scaling)

- Standard Deviation (σ): After calculation, the standard deviation is approximately 34,409.3.

Step 2: Apply the formula to each 'Income' value.

- Person A (50,000):

$$\frac{50,000 - 94,000}{34,409.3} \approx -1.279$$

Similarly, Person B (65,000): -0.843

Person C (120,000): 0.756

Person D (140,000): 1.337

Person E (95,000): 0.029

Person	Age	Income
A	25	50,000
B	30	65,000
C	45	120,000
D	50	140,000
E	35	95,000



Person	Original Income	Normalized Income	Standardized Income
A	50,000	0.000	-1.279
B	65,000	0.167	-0.843
C	120,000	0.778	0.756
D	140,000	1.000	1.337
E	95,000	0.500	0.029

Data Preprocessing : Data Transformation

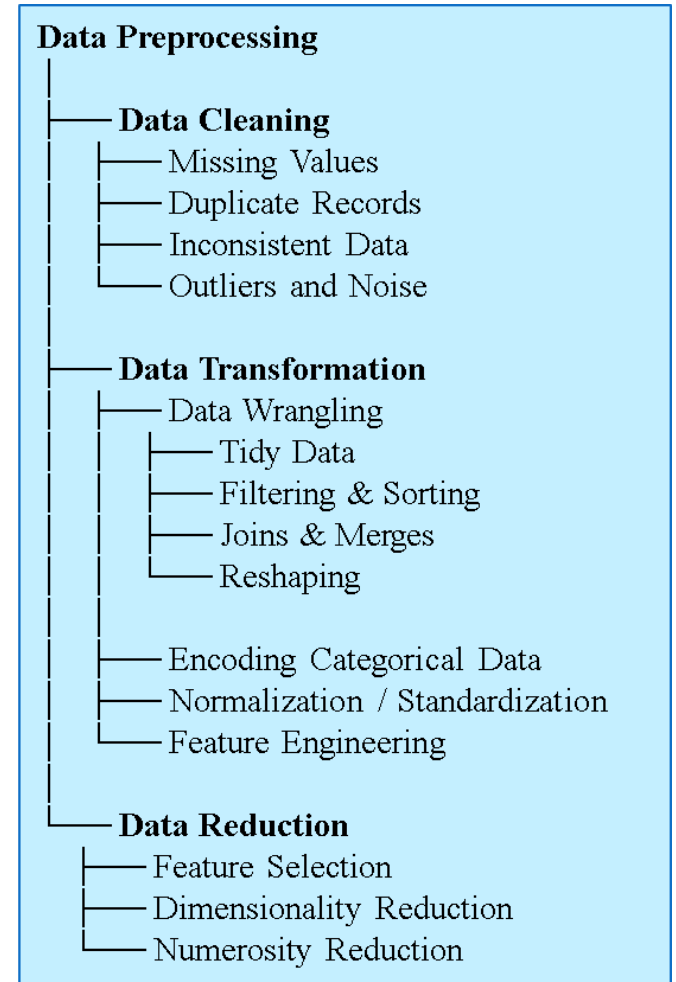
Feature Engineering

The process of creating new, informative features (variables) from existing data to improve analysis and machine learning performance.

Why Do We Need Feature Engineering?

Raw data is often not the most useful form for analysis.

Can I create a new variable that captures more useful information than the original variables?



Data Preprocessing : Data Transformation

Common Feature Engineering Tasks

1. Creating New Features

Student	Midterm	Final
Rahim	70	80
Karim	65	75



Student	Midterm	Final	Total
Rahim	70	80	150
Karim	65	75	140

The new feature **Total** may be more informative

Data Preprocessing : Data Transformation

Common Feature Engineering Tasks

2. Deriving Features from Dates

Date
2025-06-10
2025-12-25



Year	Month	Day
2025	6	10
2025	12	25

Data Preprocessing : Data Transformation

Common Feature Engineering Tasks

3. Creating Age from Birth Year

Birth Year
2003
2001

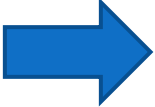


Birth Year	Age
2003	22
2001	24

Data Preprocessing : Data Transformation

Common Feature Engineering Tasks

4. Creating Categories (Binning)

Age		Age	Age Group
18		18	Young
22		22	Young
35		35	Adult
60		60	Senior

Data Preprocessing : Data Transformation

Common Feature Engineering Tasks

5. Creating Ratios

Income	Expenses
50000	30000
70000	35000



Income	Expenses	Savings Rate
50000	30000	0.40
70000	35000	0.50

Data Preprocessing : Data Transformation

Feature Engineering: Another Example

Original Features	Engineered Features
area_sqft	price_per_sqft
num_bedrooms	bedrooms_squared
build_year	age_group, house_age
location	proximity_to_center, has_good_school

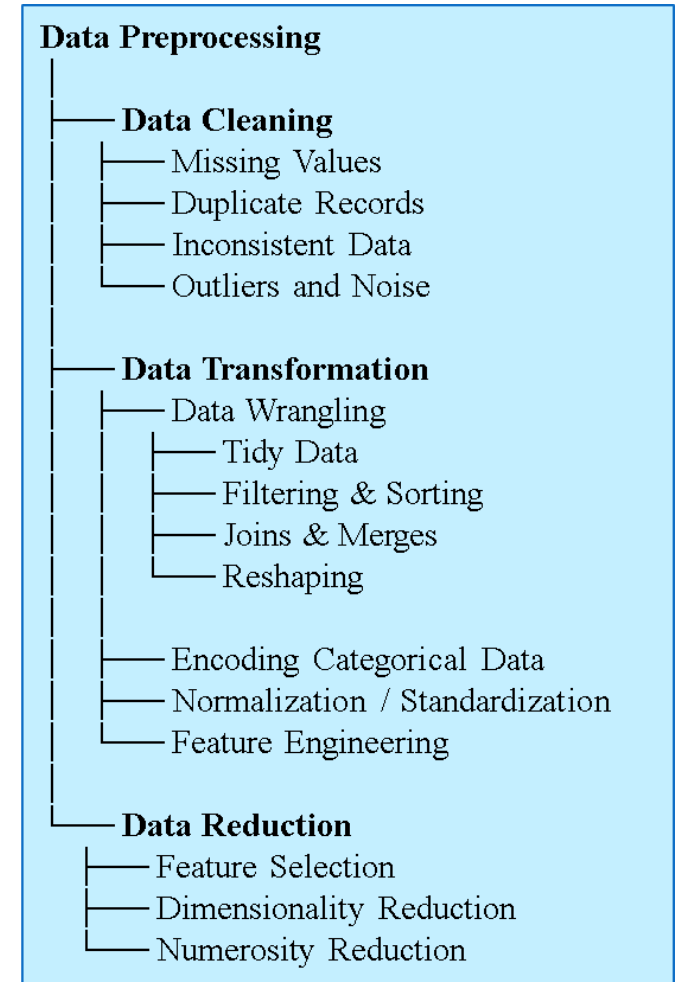
Data Preprocessing : Data Reduction

Feature Selection

Feature Selection is the process of selecting the most relevant features (variables) from a dataset and removing unnecessary ones.

Goal

- Reduce dataset complexity
- Improve model performance
- Reduce training time
- Improve interpretability



Data Preprocessing : Data Reduction

Feature Selection

Suppose we want to predict student performance.

Age	Height	Weight	Study Hours	Attendance	CGPA
20	170	65	15	85	3.5

Do we need all variables?

Maybe:

Study Hours, Attendance are more useful than:

Height, Weight for predicting CGPA.

Feature Selection helps us identify and keep the important variables.

Data Preprocessing : Data Reduction

Feature Selection vs. Feature Engineering

Feature Engineering	Feature Selection
Creates new features	Chooses existing features
Adds variables	Removes variables
Example: Total = Midterm + Final	Remove Height and Weight

How Do We Select Features?

Method 1: Domain Knowledge

Example: To predict CGPA:

✓ Study Hours ✓ Attendance ✗ Favorite Color

Data Preprocessing : Data Reduction

How Do We Select Features?

Method 2: Correlation Analysis

Compute correlation with the target variable.

Feature	Correlation with CGPA
Study Hours	0.82
Attendance	0.74
Height	0.05

Conclusion: Height may be removed.

Data Preprocessing : Data Reduction

How Do We Select Features?

Method 3: Variance Threshold

Remove features with almost no variation.

Student	Country
1	Bangladesh
2	Bangladesh
3	Bangladesh

This variable contains little information.

Data Preprocessing : Data Reduction

Dimensionality Reduction: This reduces the number of features (columns). High-dimensional data can lead to poor model performance (the "**curse of dimensionality**"). Techniques like **Principal Component Analysis** (PCA) create a smaller set of new features that capture most of the original data's variance.

Data Preprocessing

Data Cleaning

- Missing Values
- Duplicate Records
- Inconsistent Data
- Outliers and Noise

Data Transformation

- Data Wrangling
 - Tidy Data
 - Filtering & Sorting
 - Joins & Merges
 - Reshaping
- Encoding Categorical Data
- Normalization / Standardization
- Feature Engineering

Data Reduction

- Feature Selection
- Dimensionality Reduction
- Numerosity Reduction

Data Preprocessing : Data Reduction

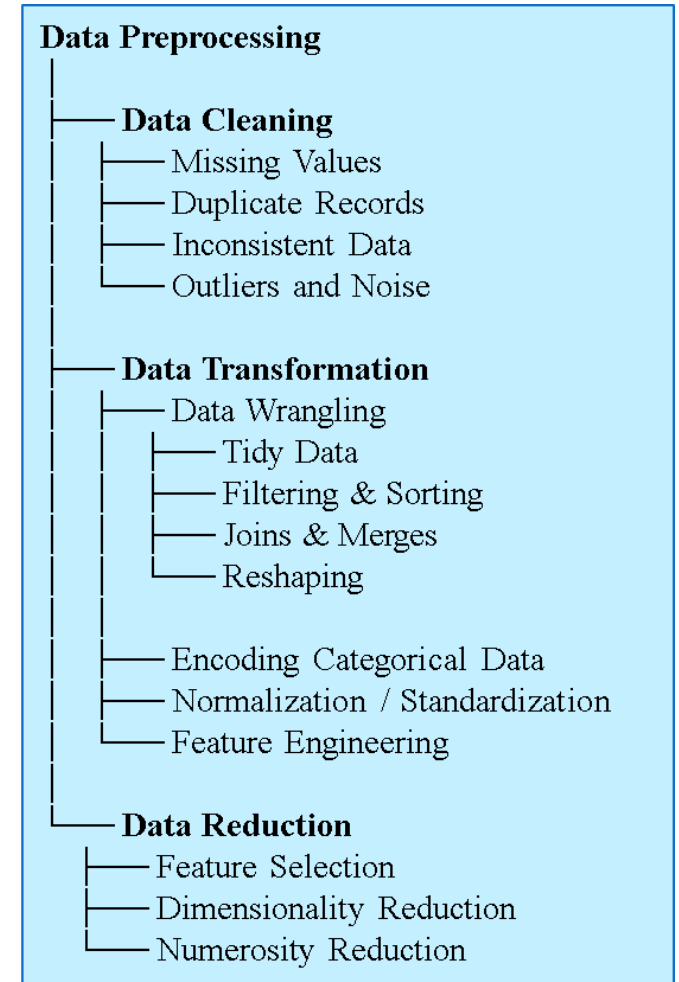
Details of PCA will be taught later

Data Preprocessing : Data Reduction

Numerosity Reduction: This **reduces the number of data entries** (rows). When dealing with massive datasets, you can use **sampling** (selecting a representative subset of the data) or use models like **clustering or regression** to represent the data more compactly.

Goal

- Reduce storage requirements
- Speed up analysis
- Reduce computational cost
- Preserve key information



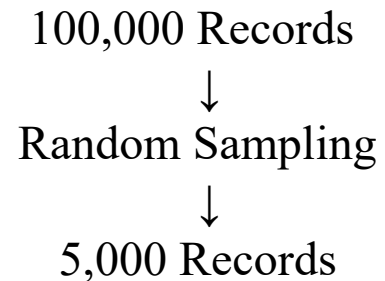
Data Preprocessing : Data Reduction

Types of Numerosity Reduction

1. Sampling

Suppose a company has:10,000,000 customer records. Do we always need to analyze all 10 million records?

Sometimes we can use a smaller representation and obtain similar insights.



Data Preprocessing : Data Reduction

Types of Numerosity Reduction

2. Aggregation

Combine multiple records into summarized records.

Example: Daily Sales

Day	Sales
Mon	100
Tue	120
Wed	110
Thu	130
Fri	140

Aggregate to weekly sales:

Week	Total Sales
Week 1	600

Data Preprocessing : Data Reduction

Types of Numerosity Reduction

3. Clustering-Based Reduction

Group similar observations together and represent each group by its centroid.

Original Data

Customer 1 → Income = 50K

Customer 2 → Income = 52K

Customer 3 → Income = 49K

Representative: Cluster Center = 50K

Instead of storing every record, store the cluster representative.

Data Preprocessing : Data Reduction

Types of Numerosity Reduction

4. Histograms / Binning

Store value ranges rather than individual observations.

Original Data

20, 21, 22, 23, 24, 25

Reduced Form

Age Range	Frequency
20–22	3
23–25	3

Instead of storing every record, store the cluster representative.

Data Preprocessing : Data Reduction

Numerosity Reduction vs Dimensionality

Numerosity Reduction	Dimensionality Reduction
Reduces rows	Reduces columns
Fewer observations	Fewer features
Example: Sampling	Example: PCA